
MeanFlow-Transfer: Joint Adaptation and Distillation for Any Diffusion and Flow Model

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Abstract

Adapting pretrained diffusion and flow models to new domains while enabling fast generation remains a challenging problem. SoTA approaches typically treat adaptation and distillation as separate stages, requiring complex design. Furthermore, they remain tied to the source model’s parameterization, limiting the ability to leverage more efficient sampling formulations across model families. We propose **MeanFlow-Transfer (MeFT)**, a unified framework that performs joint adaptation and distillation on any diffusion or flow-based pretrained generative model. The resulting model is MeFT bridges differences in source parameterizations by mapping model outputs to a shared instant-velocity representation, providing a common transfer space for models that predict noise, data, velocity, or mean velocity. It then optimizes a MeanFlow objective on the target domain, enabling adaptation and few-step distillation within a single training loop. To improve target-domain alignment under limited data, MeFT builds on DogFit [1] and extends its source-anchored guidance to heterogeneous source models by mapping source predictions to instantaneous velocity and aligning the source noise schedule with the target MeanFlow. Using MeFT, we adapt ImageNet-pretrained transformer-based models, including DiT, SiT, JiT, and iMF, to five target domains. Results indicate that MeFT outperforms standard fine-tuning for few-step generation across all source models and target domains on average. This establishes MeFT as a universal transfer framework with a strong sampling speed–quality trade-off.

Our code is included in the supplementary materials and will be made public.

1 Introduction

Pretrained diffusion and flow-based generators provide strong generic priors for image synthesis, making them attractive source models for transfer to new domains [2–6, 1, 7, 8]. Yet, adapting these models typically preserves the costly multi-step sampling procedure of the source generator. This limits their practical usefulness for personalized applications that require fast, low-cost and interactive generation. In parallel, few-step generation has advanced rapidly, but these methods are usually developed as standalone generation paradigms rather than as transfer recipes for adapting pretrained source models [9–13]. As a result, efficient generation and target-domain adaptation have largely been treated as separate problems.

Recent work has started reducing this gap. Uni-DAD shows that adaptation and distillation can be unified within a single-stage pipeline for diffusion models, avoiding the complexity of adapt-then-distill or distill-then-adapt designs [8]. However, a key obstacle to universal transfer remains: pretrained source generators are not expressed in a common prediction space. In practice, modern source models may predict the input x , noise ϵ , instantaneous velocity v , or mean velocity u , and this heterogeneity makes it difficult to design a single adaptation procedure that remains valid across model families.

At the same time, the MeanFlow family suggests a promising direction for fast generation [13, 16, 17]. By modeling average velocity rather than instant velocity, MeanFlow enables efficient one-step or few-step sampling. However, existing MeanFlow methods focus on training within a fixed parameterization and domain, and do not address (i) transfer from heterogeneous source prediction spaces (e.g., x , ϵ , v , u), or (ii) joint adaptation and distillation within a unified framework.

In this work, we propose MeanFlow-Transfer (MeFT), a universal framework for jointly adapting and distilling pretrained diffusion and flow models. Our key idea is to use instant velocity as a common transfer space and optimize a MeanFlow objective in the target domain. MeFT operates in two steps. First, it maps the output of a pretrained source model — regardless of whether it predicts x , ϵ , v , or u — into a shared instant velocity representation, enabling consistent initialization of a MeanFlow model. Second, it fine-tunes this model on target-domain data using an iMF-style objective with source-anchored guidance. Specifically, the guidance offset uses the fixed source model as the unconditional anchor, rather than relying on an under-trained target unconditional branch, leading to more stable and effective adaptation in low-data regimes [1].

This formulation turns MeanFlow from a standalone fast generation method into a unified transfer framework for pretrained generative models. We evaluate MeFT using ImageNet-pretrained DiT, SiT, JiT, and iMF models across five target domains. Across all source models, MeFT consistently outperforms standard fine-tuning in the few-step regime, achieving improved generation quality while maintaining low inference cost. These results position MeFT as a practical and universal transfer framework for efficient few-step generative modeling.

Contributions. (i) We cast source-to-target adaptation under heterogeneous parameterizations as transfer into a few-step MeanFlow generator, with a unified initialization that recovers instantaneous velocity from x -, ϵ -, v -, and u -predictors. (ii) We introduce a domain-guided iMF objective with source-anchored guidance that enables stable target adaptation and few-step distillation in a single training loop. (iii) We show that MeFT enables consistent few-step generation gains over standard fine-tuning across five source models and five target domains.

2 Related Work

Diffusion Acceleration. Recent work has devoted substantial effort to accelerating the sampling speed of diffusion models and their flow variants. Distillation-based approaches compress a pretrained many-step sampler into a one-step or few-step generator while preserving sample quality, for example through progressive [9], distribution-matching [11], or adversarial [12] distillation. Consistency Models [10] instead learn few-step generators from scratch by enforcing consistency along denoising trajectories. This idea has been further extended to *flow maps*, which model transitions between arbitrary timestep pairs [18–20]. MeanFlow is one such formulation, parameterizing the *average velocity* between timestep pairs [13]. Subsequent work improves this framework in different ways: Improved MeanFlow (iMF) reformulates the training objective for greater stability [16], while Decoupled MeanFlow adapts pretrained flow-matching models into MeanFlow-style generators by conditioning late network blocks on a second timestep [17]. These models, after distillation, remain confined to their teacher or initial training domain. Moreover, distillation procedures are often tightly coupled to the architecture and prediction parameterization, limiting their flexibility in model-agnostic domain transfer.

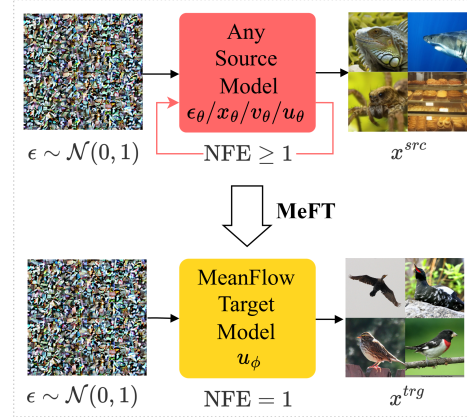


Figure 1: MeFT distills a diffusion or flow model with any number of function evaluations (NFE) and any parameterization into a MeanFlow model while adapting it to a new domain. Source and target domains: ImageNet [14] and Birds [15].

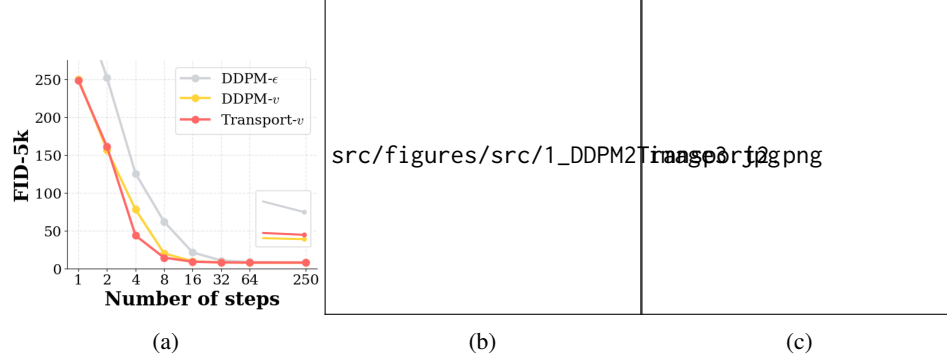


Figure 2: Challenges in joint adaptation and distillation addressed by MeFT. (a) Initial model is an ImageNet pretrained DiT with DDPM- ϵ state.

Diffusion Adaptation. Pretrained diffusion models exhibit strong transferability, motivating extensive work on adapting them to new domains under limited data. Rather than training from scratch, these methods adapt a pretrained source model while preserving its priors, controlling how source-domain information is forgotten, retained, or re-injected during transfer [2, 21, 7, 6]. Recent progress emphasizes *data-efficient* transfer for unconditional [22, 3, 23] and subject-driven [24] generation, as well as *parameter-efficient fine-tuning* (PEFT) [4, 25, 5]. However, most approaches still rely on multi-step diffusion sampling at inference. Optimizing the *sampling cost* of adapted generators, therefore, remains an important challenge. Despite this progress, existing methods remain limited in generality, often tied to specific distillation pipelines, guidance mechanisms, or source model families. More broadly, jointly achieving *fast sampling*, *high target-domain quality*, and *broad compatibility across pretrained diffusion and flow-based generators* remains challenging. This motivates transfer frameworks that can convert diverse pretrained models into efficient target-domain generators within a unified formulation.

Diffusion Adaptation and Acceleration. Recent work explores the intersection of *adaptation* and *acceleration*, aiming to obtain generators that are both aligned with the target distribution and efficient at inference. A key challenge is adapting *already distilled* generators. Pairwise Sample Optimization [26] shows that tuning few-step distilled diffusion models requires objectives beyond naive fine-tuning. In a similar spirit, some works study whether distilled generators remain compatible with lightweight adaptation modules such as LoRA [27, 28], while others introduce plug-and-play acceleration modules applicable even after adaptation [29, 30]. A more integrated direction jointly addresses *distillation and adaptation*. Uni-DAD [8], for example, unifies these objectives within a single framework for few-step few-shot generation, directly targeting fast target-domain synthesis. Related approaches couple diffusion distillation with downstream objectives such as image conditioning [31], machine unlearning [32], and reinforcement learning [33]. Closely related, DogFit [1] emphasizes domain-guided fine-tuning for efficient transfer while preserving generation quality. A more integrated direction jointly addresses *distillation and adaptation*. Despite this progress, existing methods remain limited in generality, often tied to specific distillation pipelines, guidance mechanisms, or source model families. More broadly, jointly achieving *fast sampling*, *high target-domain quality*, and *broad compatibility across pretrained diffusion and flow-based generators* remains challenging. This motivates transfer frameworks that can convert diverse pretrained models into efficient target-domain generators within a unified formulation.

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From noise prediction to a common velocity interface. Diffusion models are typically sampled using the same noise schedule and prediction target used during training. We revisit this assumption using the ImageNet DiT model [34], which is trained with a DDPM noise schedule [35] to predict noise ϵ ; we refer to this native setting as DDPM- ϵ . Starting from the same frozen model, we first keep the DDPM trajectory fixed and convert the predicted noise into the instantaneous velocity induced by that trajectory, yielding DDPM- v . This changes only the prediction parameterization while preserving the original schedule, yet Fig. 2a shows that it recovers the few-step sampling efficiency of velocity/flow-based models without retraining or finetuning. We next consider Transport- v , which

maps the DDPM trajectory onto a linear transport flow-matching path [36–38] through explicit time and state alignment, following Diff2Flow [39]. This further changes the underlying probability path, yet preserves the few-step gains after alignment. These results motivate our universal velocity transfer setting: pretrained generators, regardless of their original schedule or prediction target, can be mapped into a velocity interface over a new probability path. This enables the transfer of a DDPM prior into target velocity or MeanFlow models defined on linear transport trajectories. Detailed formulations are provided in Appx. B.1.

Learning velocity mapping or remaining in original space with explicit velocity mapping. It has been shown in Diff2Flow, that it is better to apply transformations and alignment, not naively force DDPM model to learn everything by itself when finetuning a model on the same source dataset. This is especially true when using LoRA and PEFT and limited compute. We also investigate whether this holds true in transfer learning (Limited data - few-shot data - domain-shift)

Rather than naively fine-tuning a DDPM model with a velocity objective, we explicitly align the source trajectory to the target velocity space. This avoids forcing the pretrained model to simultaneously learn a new output parameterization, a new time convention, and a new probability path. The source model can therefore preserve its pretrained DDPM prior while providing velocity-compatible supervision for the target MeanFlow model.

Compared to Diff2Flow, we first use the timestep and state alignment for meanflow instead of flow matching. Further, we show the effect in transfer learning which is a significantly different task than fine-tuning on the same data.

Further, we notice that if we do not find a way to transport the source model into a velocity transport path, it will not be possible to use DogFit or Domain Guidance as the source and target models live in different spaces.

We occlude time-step transformation in our figures and math, but bear in mind that often times, the time-step needs to be transformed from a source to a target model. For example, DiT timesteps are from 999 to 0 while SiT is from 0 to 1. When initializing from DiT, we flip and multiply the time-steps so the model is not shocked by different scale of inputs.

3 Proposed MeanFlow-Transfer Method

This paper addresses the source-to-target adaptation of pretrained generative models under heterogeneous source parameterizations. A pretrained *source generator* with parameters θ may take the form of an x -, ϵ -, v -, or u -predictor, generating samples from a source distribution $x^{\text{src}} \sim p(x^{\text{src}})$. Given limited target-domain data, our goal is to jointly adapt the model to the target distribution $x^{\text{trg}} \sim p(x^{\text{trg}})$ while distilling it into a few-step MF *target generator* u_ϕ with parameters ϕ . This yields high-quality, target-aligned generation while avoiding the costly multi-step sampling required by x -, ϵ -, and v -parameterized models.

We assume that the target model shares a similar structure to the source model, enabling the transfer of weights and initialization. We further assume access to the source model’s weights for initialization and having access to a copy of the source model during training. While source models may have different parameterizations and noise schedules (interpolation paths), the goal is to unify these variations under a common framework. Our framework applies to both conditional and unconditional target-domain generation and is not restricted to a single conditioning setting.

3.1 Background

Flow Matching (FM) FM [37, 36, 40] learns a time-dependent velocity field that transports samples from a simple prior distribution to the data distribution. Let z_t denote an intermediate state along a probability path between data $x \sim p_{\text{data}}$ and Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$. The FM objective targets the *marginal velocity* field:

$$v(z, t) = \mathbb{E} \left[\frac{dz_t}{dt} \middle| z_t = z \right], \quad (1)$$

177 In practice, this marginal velocity is not directly accessible. For the common linear interpolation path
 178 $z_t = (1 - t)x + t\epsilon$, the corresponding *conditional velocity* is $v^*(x, \epsilon, t) = \frac{dz_t}{dt} = \epsilon - x$. FM trains a
 179 neural network v_θ using the regression objective:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t,x,\epsilon} \left[\|v_\theta(z_t, t) - v^*(x, \epsilon, t)\|^2 \right]. \quad (2)$$

180 Although the loss uses the conditional target $v^*(x, \epsilon, t)$, the minimizer of this mean-squared objective
 181 is the conditional expectation $\mathbb{E}[v^*(x, \epsilon, t) \mid z_t = z]$, which recovers the marginal velocity field
 182 $v(z, t)$ [37]. When clear, we omit t and write $v(z_t)$ instead of $v(z_t, t)$.

183 **MeanFlow (MF)** While FM learns the *instant velocity* field $v(z_t)$, MF [13] instead considers the
 184 *average velocity* between two time steps r and t . It defines the *average velocity* field as:

$$u(z_t, r, t) \triangleq \frac{1}{t - r} \int_r^t v(z_\tau) d\tau. \quad (3)$$

185 When clear from context, we omit the explicit dependence on (r, t) and write $u(z_t)$. Directly
 186 supervising the integral above is intractable during training. To avoid this, MF derives an identity that
 187 links the *instant velocity* to the *mean velocity* through differentiation with respect to t :

$$u(z_t) = v(z_t) - (t - r) \frac{d}{dt} u(z_t). \quad (4)$$

188 where in practice, $du(z_t)/dt = v(z_t, t)\partial_z u + \partial_t u$ is computed through a Jacobian–vector product
 189 (JVP) [13]. This relation, referred to as the *MF identity* [13], allows the original velocity regression
 190 objective to be rewritten in terms of a u -predictor NN. Using this identity, improved MF (iMF) [16]
 191 defines a *compound function* V_θ :

$$V_\theta(z_t) \triangleq u_\theta(z_t) + (t - r) \text{JVP}_{\text{sg}}(u_\theta; v_\theta). \quad (5)$$

192 where sg denotes stop-gradient. To allow classifier-free guidance (CFG) [41] at sampling time, iMF
 193 regresses a *guided target velocity* v_g^* [42]:

$$v_g^* = v^* + w(v_\theta(z_t, c) - v_\theta(z_t, \emptyset)), \quad (6)$$

194 where c and $\emptyset$ denote the presence and absence of a condition (e.g., class label, text), respectively,
 195 and w is the guidance scale that can either be fixed or sampled and given as an additional input
 196 condition to the model to allow flexible guidance at test-time. Then, the *mean velocity* predictor
 197 and *compound function* become conditioned on both c and w , yielding $u_\theta(z_t, c, w)$ and $V_\theta(z_t, c, w)$.
 198 The *instant velocity* v_θ may either be recovered from the *mean velocity* via $v_\theta(z_t, t) = u_\theta(z_t, t, t)$ or
 199 predicted by a separate v -head [16]. The model is then trained with an FM-like regression objective:

$$\mathcal{L}_{\text{iMF}} = \mathbb{E} \left[\|V_\theta(z_t, c) - v_g^*\|^2 \right]. \quad (7)$$

200 This formulation can be viewed as u -prediction with v -loss. After training, iMF supports few-step
 201 sampling by evaluating u_θ over a coarse sequence of time intervals. Specifically, one-step sampling
 202 is achieved via $x = \epsilon - u_\theta(\epsilon, 0, 1, c, w)$.

203 3.2 MeanFlow-Transfer

204 We propose **MeanFlow-Transfer (MeFT)**, a universal framework for joint target-domain adaptation
 205 and flow distillation from any source generator. At a high level, MeFT converts heterogeneous source
 206 generators into a unified instant velocity representation, uses it to initialize a MF model, and then
 207 adapts it to the target domain using a source-anchored training guidance. MeFT consists of two key
 208 components. First, a *universal instant velocity mapping* transforms different source parameterizations
 209 into a shared velocity representation, enabling consistent MF initialization. Second, *domain-guided*
 210 *fine-tuning* adapts the initialized model to the target domain using an iMF-style objective with an
 211 improved guidance mechanism for stronger target alignment.

212 **Universal instant velocity mapping.** Pretrained source generators operate in different prediction
 213 spaces $(x, \epsilon, v, \text{ or } u)$, which prevents direct MF distillation within a unified framework. MeFT resolves
 214 this mismatch by recovering a shared *instant velocity* representation from each parameterization.

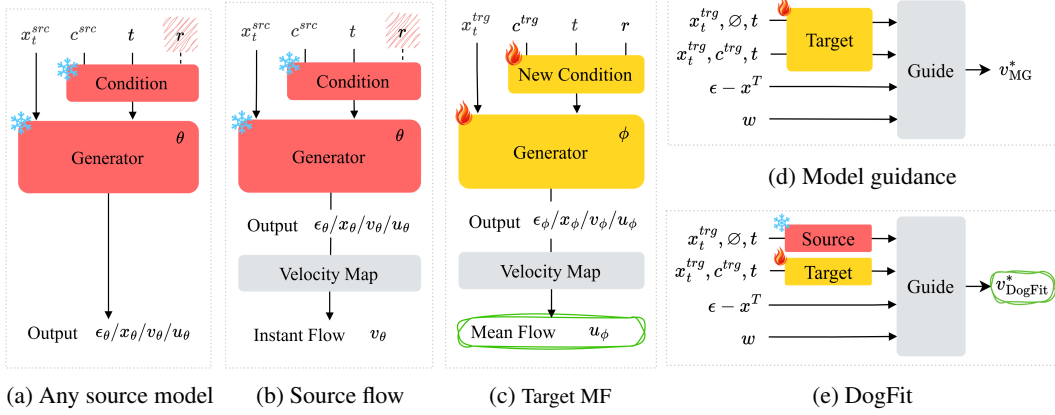


Figure 3: (a) A pretrained source generator may predict in different spaces (x , ϵ , v , or u). (b) MeFT first maps the source model’s output into a common instant velocity representation. (c) This enables initialization of a target MF model for training, with the additional condition r added when it is not already present. (d) Standard model guidance uses the target model for both conditional and unconditional instant velocity prediction. (e) MeFT uses source-anchored guidance (DogFit), where the fixed source model provides the unconditional anchor during target-domain fine-tuning.

Table 1: Universal MeFT initialization from any source parameterization. Under the linear interpolation path, MeFT recovers a consistent instant velocity from each prediction space and uses it to initialize the target MeanFlow model.

Prediction space	$\mathbf{x}$	ϵ	$\mathbf{v}$	$\mathbf{u}$
Source Prediction	$x_\theta(z_t)$	$\epsilon_\theta(z_t)$	$v_\theta(z_t)$	$u_\theta(z_t), v_\theta(z_t)^\dagger$
Recovered instant velocity v_θ	$\frac{z_t - x_\theta(z_t)}{t}$	$\frac{\epsilon_\theta(z_t) - z_t}{1 - t}$	$v_\theta(z_t)$	$v_\theta(z_t)^\dagger$
Mean velocity initialization u_ϕ	$u_\phi \leftarrow v_\theta$		$u_\phi \leftarrow u_\theta, v_\phi \leftarrow v_\theta^\dagger$	

[†] We preserve the *source generator* design and do not add new heads. We use an existing velocity head when available (e.g., for iMF [16]); otherwise, we recover the *instantaneous velocity* via $v(z_t, t) = u(z_t, t, t)$.

Specifically, assuming the standard linear interpolation path $z_t = (1 - t)x + t\epsilon$, we invert the source prediction to obtain the corresponding velocity field consistently. This provides a common interface across heterogeneous models, enabling consistent initialization of a target MF generator and providing a source-model velocity used by DogFit. Table 1 summarizes the mapping for each parameterization.

Domain-guided fine-tuning (DogFit). After initialization, MeFT adapts the MF target generator to the target domain through a guided iMF-style objective. Starting from the initialized MF model, we construct the corresponding instant velocity predictor and optimize it toward a guided target velocity:

$$v_{\text{DogFit}}^* = v^* + \left(1 - \frac{1}{w}\right) (v_\phi(z_t, c) - v_\theta(z_t, \emptyset)), \quad (8)$$

where $v_\theta(z_t, \emptyset)$ is derived from the fixed source model and $v_\phi(z_t, c)$ from the target model. w can either be a fixed value or sampled and provided as an additional conditioning input to the model [1, 16]. We follow the sampling distribution and value range used in iMF [16]. Unlike standard model guidance [42], which relies on a poorly learned unconditional branch in low-data regimes, MeFT anchors the unconditional prediction to the fixed source model. This provides a stable reference during fine-tuning, prevents drift toward out-of-distribution generations, and is particularly effective when target data is limited [1, 6]. The source model v_θ is kept fixed during training.

The model is optimized with the regression loss:

$$\mathcal{L}_{\text{MeFT}} = \mathbb{E} \left[\|V_\phi(z_t, c, w) - v_{\text{DogFit}}^*\|^2 \right]. \quad (9)$$

Algorithm 1: MeFT initialization.

```
# src: pretrained source model
# param_type: one of {x, eps, v, u}

class VelocityWrapper(src, param_type):
    def forward(z, t, c):
        if param_type == "x":
            x_hat = src(z, t, c)
            return (z - x_hat) / t

        elif param_type == "eps":
            eps_hat = src(z, t, c)
            return (eps_hat - z) / (1 - t)

        elif param_type == "v":
            return src(z, t, c)

        elif param_type == "u":
            return src(z, t, t, c)
            # or use velocity head

src = VelocityWrapper(src, param_type)
tgt = deep_copy(src)
tgt = add_r_w_c_inputs(tgt)
```

Algorithm 2: MeFT domain-guided training.

```
# src: velocity-wrapped source model
# tgt: target MeanFlow model
# x: target-domain training batch
# c: condition batch

t, r, w = sample_t_r_cfg()
e = randn_like(x)
v_star = e - x

z = (1 - t) * x + t * e

v_c = tgt(z, t, t, w, c)
v_u = src(z, t, None)

v_g = v_star + (1 - 1/w) * (v_c - v_u) Eq. (8)

u, dudt = jvp(tgt, (z, r, t, w, c),
               (v_c, 0, 1, 0, 0))

V = u + (t - r) * stopgrad(dudt)
error = V - stopgrad(v_g)

loss = metric(error)
```

230 In practice, gradients are stopped through v_{DogFit}^* . Otherwise, the target would depend on the trainable
231 prediction $v_\phi(z_t, c)$, creating a self-referential objective that can lead to degenerate optimization.

232 Algorithm 1 and 2 summarize MeFT initialization and domain-guided MF training, respectively.

233 4 Results and Discussion

234 4.1 Experimental Setup

235 As the source model, we use pretrained ImageNet generators including DiT/XL-2 [34], SiT/XL-2 [?],
236 JiT/XL-2 [43], iMF/XL-2 [16], and pMF/XL-2 [44]. These models cover x -, ϵ -, v -, and u -prediction
237 settings, allowing us to evaluate MeFT under a wide range of adaptation recipes. We adapt each source
238 model using MeFT to a diverse collection of labeled target domains: ArtBench [45], Caltech [46],
239 CUB-Birds [47], Food [48], and Stanford-Cars [15]. This setup is designed to span diverse semantic
240 domains and domain-shift magnitudes, in the spirit of prior transfer-learning studies [1, 6, 5]. We
241 compare MeFT against standard fine-tuning (FT) initialized from the same pretrained generator.
242 At inference, we report 1-, 4-, and 250-step generation. We evaluate generation quality using FID,
243 $\text{FD}_{\text{DINOv2}}$ (FDD), and IS using 10K generated samples and the complete target datasets. Unless
244 otherwise stated, all models use 256×256 image resolution. All models are initialized from backbones
245 pretrained on ImageNet at 256×256 resolution. The model variants we use are DiT-XL/2, SiT-XL/2,
246 JiT-H-16, and iMF-XL-2. See Table 4 for more details on hyperparameters and each backbone.

247 4.2 Main Results

248 4.3 Ablations

249 [Guidance Interval](#). [EMA](#). [DogFit](#). [Straight change of parameter vs only change loss trajectory length](#)
250 [vs adaptation Speed of adaptation of different trajectory lengths](#)

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Table 2: FID, FD_{DINOv2} (FDD), and Inception Score (IS) comparison across labeled target domains. We report quality and the number of function evaluations (NFE) at inference.

Initialization	Method	NFE	FID					
			ArtBench	Caltech	CUB-Birds	Food	Stanford-Cars	Avg.
DiT/XL-2	FT	250×2	11.90	24.47	7.75	9.26	5.95	11.87
	FT	4×2	194.80	123.03	147.18	199.16	196.13	172.06
	MeFT	4	—	—	—	—	—	—
SiT/XL-2	FT	250×2	10.21	24.20	4.19	6.81	6.93	10.47
	FT	4×2	36.11	33.26	14.33	22.22	34.83	28.15
	MeFT	4	15.66	27.13	4.78	12.81	6.78	13.43
JiT/XL-2	FT	50×2	—	—	—	—	—	—
	FT	4×2	—	—	—	—	—	—
	MeFT	4	—	—	—	—	—	—

Initialization	Method	NFE	FDD					
			ArtBench	Caltech	CUB-Birds	Food	Stanford-Cars	Avg.
DiT/XL-2	FT	250×2	211.02	394.66	167.75	277.95	164.10	243.10
	FT	4×2	1726.58	1309.74	1874.65	2295.54	2053.23	1851.95
	MeFT	4	—	—	—	—	—	—
SiT/XL-2	FT	250×2	205.05	399.26	111.66	343.67	185.82	249.09
	FT	4×2	457.69	543.04	310.21	620.82	624.10	511.17
	MeFT	4	298.58	489.50	215.25	562.73	265.90	366.39
JiT/XL-2	FT	50×2	—	—	—	—	—	—
	FT	4×2	—	—	—	—	—	—
	MeFT	4	—	—	—	—	—	—

Initialization	Method	NFE	IS					
			ArtBench	Caltech	CUB-Birds	Food	Stanford-Cars	Avg.
DiT/XL-2	FT	250×2	7.15	34.91	6.52	5.96	3.45	11.60
	FT	4×2	3.10	12.79	6.12	4.77	5.18	6.39
	MeFT	4	—	—	—	—	—	—
SiT/XL-2	FT	250×2	6.94	35.23	6.03	5.45	3.39	11.41
	FT	4×2	5.45	31.99	6.96	5.35	3.52	10.65
	MeFT	4	5.91	28.54	5.59	4.63	3.02	9.54
JiT/XL-2	FT	50×2	—	—	—	—	—	—
	FT	4×2	—	—	—	—	—	—
	MeFT	4	—	—	—	—	—	—

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Table 3: FID and FD_{DINOv2} comparison across labeled target domains for MeanFlow-family teacher initializations. We report quality and the number of function evaluations (NFE) at inference.

Initialization	Method	NFE	FID					
			ArtBench	Caltech	CUB-Birds	Food	Stanford-Cars	Avg.
iMF/XL-2 [16] (<i>v</i> -loss, <i>u</i> -pred)	FT	4	10.60	27.54	7.26	10.58	9.69	13.13
	FT	1	11.21	34.92	11.74	12.20	15.41	17.10
	MeFT	4						
	MeFT	1						
Initialization	Method	NFE	FD_{DINOv2}					
			ArtBench	Caltech	CUB-Birds	Food	Stanford-Cars	Avg.
iMF/XL-2 [16] (<i>v</i> -loss, <i>u</i> -pred)	FT	4	273.28	471.09	310.62	465.38	276.53	359.38
	FT	1	268.85	611.91	423.50	521.62	403.39	445.85
	MeFT	4						
	MeFT	1						
Initialization	Method	NFE	IS					
			ArtBench	Caltech	CUB-Birds	Food	Stanford-Cars	Avg.
iMF/XL-2 [16] (<i>v</i> -loss, <i>u</i> -pred)	FT	4	7.36	31.46	6.08	6.37	3.41	10.94
	FT	1	7.70	28.43	6.37	6.31	3.43	10.45
	MeFT	4						
	MeFT	1						

height

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A Appendix / supplemental material

B Additional Explanations

B.1 Diffusion to Flow-matching

Using the common diffusion interpolant:

$$x_{\tau}^{\text{DM}} = \alpha_{\tau}x_0 + \sigma_{\tau}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (10)$$

we define three ways of sampling from a pretrained DDPM model. We refer to these as DDPM- ϵ , DDPM- v , and Transport- v .

DDPM- ϵ : This is the original diffusion sampler. The model predicts noise, and sampling follows the discrete reverse diffusion chain. We have

$$\hat{\epsilon}_{\theta} = \hat{\epsilon}_{\theta}(x_{\tau}, \tau), \quad \hat{x}_0 = \frac{x_{\tau} - \sigma_{\tau}\hat{\epsilon}_{\theta}}{\alpha_{\tau}}. \quad (11)$$

Sampling is the usual stochastic DDPM reverse step:

$$x_{\tau-\Delta\tau} = \mu_{\theta}(x_{\tau}, \tau) + \Sigma_{\theta}(x_{\tau}, \tau)^{1/2}z, \quad z \sim \mathcal{N}(0, I). \quad (12)$$

Table 4: ImageNet 256×256 reference performance and configuration of the backbones used in our study. For DiT and SiT, the standard reported results use 250 denoising steps with classifier-free guidance (CFG), corresponding to an effective NFE of 250×2 . JiT-H/16 uses x -prediction. pMF-H/16 decouples prediction and loss spaces, using x -prediction with a MeanFlow loss in velocity space.

	DiT-XL/2	SiT-XL/2	JiT-H/16	iMF-XL/2	pMF-H/16
<i>Source Model</i>					
Params	675M + 49M	675M + 49M	953M	610M	956M
NFE	250×2	250×2	50×2	1 / 2	1
Reference FID	2.27	2.06	1.86	1.72 / 1.54	2.22
Prediction space	ϵ	v	x	v	x
Loss space	ϵ	v	v	v	v
<i>Global Params</i>					
Image resolution	256×256	256×256	256×256	256×256	256×256
Training iteration	40K	40K	40K	40K	40K
Batch size	32	32	32	32	32
Learning rate	1e-4	1e-4	5e-5 ? 6e-6	1e-4	?
<i>FT</i>					
w / w rescaled	1.5	1.5	2.2	7.5 / 0.85	7 / 0.85
ω range	[0.0, 1.0]	[0.0, 1.0]	[0.1, 1.0]	[0.45, 0.6]	[0.2, 0.6]
<i>DMF-FT</i>					
w / w rescaled	0.6	0.6	-	-	-
ω range	[0.0, 7.0]	[0.0, 7.0]	-	-	-
DMF depth	20	20	-	-	-
<i>MeFT</i>					
w / w rescaled	0.6	0.6	?	7.5 / 0.85	7 / 0.85
ω range	[0.0, 7.0]	[0.0, 7.0]	[0.1, 1.0]	[0.45, 0.6]	[0.2, 0.6]

384 **DDPM- v** : This is still an epsilon model converted into diffusion-path velocity at sampling time. We
385 have

$$v_\tau^{\text{DM}} = \frac{dx_\tau^{\text{DM}}}{d\tau} = \dot{\alpha}_\tau x_0 + \dot{\sigma}_\tau \epsilon. \quad (13)$$

386 using the same epsilon model:

$$\hat{x}_0 = \frac{x_\tau - \sigma_\tau \hat{\epsilon}_\theta}{\alpha_\tau}, \quad \hat{v}_\tau^{\text{DM}} \approx \dot{\alpha}_\tau \hat{x}_0 + \dot{\sigma}_\tau \hat{\epsilon}_\theta. \quad (14)$$

387 That is the finite-difference version:

$$\dot{\alpha}_\tau \approx \frac{\alpha_{\tau'} - \alpha_\tau}{\tau' - \tau}, \quad \dot{\sigma}_\tau \approx \frac{\sigma_{\tau'} - \sigma_\tau}{\tau' - \tau}. \quad (15)$$

388 Then Euler-update is:

$$x_{\tau'} = x_\tau + (\tau' - \tau) \hat{v}_\tau^{\text{DM}}. \quad (16)$$

389 **Transport- v** : This is the same epsilon prior again, but now sampled on a linear flow-matching path
390 after Diff2Flow-style time and state alignment. We use the flow-matching style linear path:

$$x_t^{\text{FM}} = tx_0 + (1-t)\epsilon, \quad u_t = \frac{dx_t^{\text{FM}}}{dt} = x_0 - \epsilon. \quad (17)$$

391 Using Diff2Flow-style alignment:

$$t = \frac{\alpha_\tau}{\alpha_\tau + \sigma_\tau}, \quad x_\tau^{\text{DM}} = (\alpha_\tau + \sigma_\tau) x_t^{\text{FM}}. \quad (18)$$

Table 5: FID comparison across labeled target domains. We report quality and NFE at inference.

Initialization	Method	NFE	FID					
			ArtBench	Caltech	CUB-Birds	Food	Stanford-Cars	Avg.
DiT/XL-2	FT	250×2	11.90	24.47	7.75	9.26	5.95	11.87
	FT	4×2	194.80	123.03	147.18	199.16	196.13	172.06
	FT	2×2	350.42	263.77	323.35	330.78	329.20	319.50
	FT	1×2	432.48	371.61	349.85	388.15	383.29	385.08
	DMF+FT	250						
	DMF+FT	4						
	DMF+FT	2						
	DMF+FT	1						
	FT	250×2	10.21	24.20	6.81	6.93	10.47	height
	FT	4×2	36.11	33.26	14.33	22.22	34.83	28.15
JiT/XL-2	FT	2×2	204.08	106.96	112.81	114.03	263.52	160.28
	FT	1×2	438.27	293.38	304.89	326.87	338.38	340.36
	DMF+FT	250		26.49				
	DMF+FT	4		25.85				
	DMF+FT	2		27.52				
	DMF+FT	1		38.17				
	MeFT	250	16.75	26.37	4.14	13.98	5.36	13.32
	MeFT	4	15.66	27.13	4.78	12.81	6.78	13.43
	MeFT	2	17.91	30.44	6.00	15.75	9.42	15.90
	MeFT	1	26.22	49.27	11.20	32.98	21.21	28.18
JiT/XL-2	FT	50×2	–	–	–	–	–	–
	FT	4×2	–	–	–	–	–	–
	FT	2×2	–	–	–	–	–	–
	FT	1×2	–	–	–	–	–	–

392 Then we query the epsilon model on the aligned diffusion state/time and convert:

$$\hat{x}_0 = \frac{x_\tau^{\text{DM}} - \sigma_\tau \hat{\epsilon}_\theta}{\alpha_\tau}, \quad \hat{u}_t = \hat{x}_0 - \hat{\epsilon}_\theta. \quad (19)$$

393 Euler sampling:

$$x_{t+\Delta t}^{\text{FM}} = x_t^{\text{FM}} + \Delta t \hat{u}_t. \quad (20)$$

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Table 6: FD_{DINOv2} (FDD) comparison across labeled target domains. We report quality and the NFE at inference.

Initialization	Method	NFE	FDD					
			ArtBench	Caltech	CUB-Birds	Food	Stanford-Cars	Avg.
DiT/XL-2	FT	250×2	211.02	394.66	167.75	277.95	164.10	243.10
	FT	4×2	1726.58	1309.74	1874.65	2295.54	2053.23	1851.95
	FT	2×2	2901.15	3060.22	3513.80	3225.86	3531.46	3246.50
	FT	1×2	3080.86	3387.19	3727.62	3541.96	3784.49	3504.42
SiT/XL-2	FT	250×2	205.05	399.26	111.66	343.67	185.82	249.09
	FT	4×2	457.69	543.04	310.21	620.82	624.10	511.17
	FT	2×2	1478.72	1260.77	1487.66	1881.62	2339.97	1689.75
	FT	1×2	3087.06	3004.55	3480.02	2970.62	3573.72	3223.19
	DMF+FT	250		445.91				
	DMF+FT	4		444.49				
	DMF+FT	2		484.86				
	DMF+FT	1		689.00				
	MeFT	250	280.08	424.32	164.69	523.77	171.84	312.94
	MeFT	4	298.58	489.50	215.25	562.73	265.90	366.39
	MeFT	2	349.70	524.81	290.12	716.52	374.10	451.05
	MeFT	1	512.56	801.83	530.21	1216.38	697.07	751.61
JiT/XL-2	FT	50×2	–	–	–	–	–	–
	FT	4×2	–	–	–	–	–	–
	FT	2×2	–	–	–	–	–	–
	FT	1×2	–	–	–	–	–	–

Table 7: Inception Score (IS) comparison across labeled target domains. We report quality and NFE at inference.

Initialization	Method	NFE	IS					
			ArtBench	Caltech	CUB-Birds	Food	Stanford-Cars	Avg.
DiT/XL-2	FT	250×2	7.15	34.91	6.52	5.96	3.45	11.60
	FT	4×2	3.10	12.79	6.12	4.77	5.18	6.39
	FT	2×2	2.29	4.03	3.86	3.46	4.02	3.53
	FT	1×2	1.28	1.28	1.28	1.28	1.28	1.28
SiT/XL-2	FT	250×2	6.94	35.23	6.03	5.45	3.39	11.41
	FT	4×2	5.45	31.99	6.96	5.35	3.52	10.65
	FT	2×2	2.98	13.66	8.58	3.82	5.99	7.01
	FT	1×2	1.94	2.75	2.49	2.15	2.03	2.27
	DMF+FT	250		35.22				
	DMF+FT	4		35.15				
	DMF+FT	2		34.78				
	DMF+FT	1		29.46				
	MeFT	250	5.88	34.07	5.48	4.62	3.11	10.63
	MeFT	4	5.91	28.54	5.59	4.63	3.02	9.54
	MeFT	2	5.43	30.79	5.72	4.73	3.02	9.94
	MeFT	1	4.73	23.37	5.60	4.77	2.85	8.26
JiT/XL-2	FT	50×2	–	–	–	–	–	–
	FT	4×2	–	–	–	–	–	–
	FT	2×2	–	–	–	–	–	–
	FT	1×2	–	–	–	–	–	–